

## 一些常见结论

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## 1 Abstract Algebra 抽象代数

**Proposition 1.1.** *Let  $S$  be a monoid. If an element  $a \in S$  satisfies that there exists  $b, c \in S$  s.t.  $ba = 1, ac = 1$ , then  $b = c$ .*

**Proposition 1.2.** *Let  $R$  be an integral domain. Then  $R = \bigcap_{\alpha} R_{\mathfrak{m}_{\alpha}}$ , where  $\mathfrak{m}_{\alpha}$  varies over all the maximal ideals of  $R$ .*

**Theorem 1.1 (Schur' Theorem).** *Let  $G$  be a group. If  $Z(G)$  is of finite index in  $G$ , then  $G' = [G, G]$  is finite.*

**Proposition 1.3.**  *$G$  is a infinite group. If every nontrivial proper subgroup of  $G$  is of finite index, then  $G$  is cyclic.*

**Proposition 1.4.**  *$G$  is a finite group,  $H \leq G$  subgroup. Show there is a complete set of left cosets being a complete set of right cosets at the same time.*

**Proposition 1.5.** *All groups of order  $pq$  are either isomorphic to  $\mathbb{Z}_{pq}$  (abelian case) or  $\mathbb{Z}_p \rtimes \mathbb{Z}_q$  (nonabelian case).*

**Proposition 1.6.**  *$G$  is a  $p$ -group. Then all maximal subgroups of  $G$  is normal and of index  $p$ .*

**Proposition 1.7.**  *$G$  is a finite group. Assume that  $v_p(G) = n$ . Then the number of subgroups of order  $p^i$ ,  $n_1$ , satisfies that  $n_1 \equiv 1 \pmod{p}$ , for  $\forall 1 \leq i \leq n$ .*

**Proposition 1.8.**  *$G$  is a nonabelian simple group of order 60. Then  $G \cong A_5$ .*

**Proposition 1.9.**  *$G$  is a finite group,  $N \trianglelefteq G$  a normal group,  $p \mid \#(N)$ . Assume that  $P$  is a Sylow  $p$ -group of  $N$ . Then  $G = N_G(P)N$ .*

## 2 Topology 拓扑

**Proposition 2.1.** *Let  $X$  be a topological space.,  $S \subseteq X$  subset. Then  $S$  is closed in  $X$  if and only if  $X$  can be covered by open subsets  $U_{\alpha}$  s.t.  $S \cap U_{\alpha}$  is closed in  $U_{\alpha}$ .*

**Proposition 2.2.** *Let  $X$  be a topological space. Then every maximal irreducible subspace of  $X$  is closed.*

**Remark 2.1.** *For convenience, we call maximal irreducible subspace irreducible component.*

**Proposition 2.3.** *Let  $X$  be a noetherian topological space. Then  $X$  has only finitely many irreducible components.*

### 2.1 Dimension theory

**Proposition 2.4.** *Let  $X$  be a irreducible topological space. Then for all nonempty open subset  $U \subseteq X$ ,  $U$  is dense.*

### 3 Commutative Algebra 交换代数

In this section, all rings are commutative.

#### 3.1 Radical

**Proposition 3.1.** *Let  $R$  be a ring. The nilradical  $\mathfrak{N}$  is the intersection of all prime ideals of  $R$ .*

**Proposition 3.2.** *Let  $R$  be a ring. Define the Jacobson radical  $\mathfrak{R}$  is the intersection of all maximal ideals of  $R$ . We have  $x \in \mathfrak{R} \iff 1 - xy$  is a unit in  $R$  for all  $y \in R$ .*

**Proposition 3.3.** *Let  $R$  be a ring,  $I, J \subseteq R$  ideals. If  $\sqrt{I}$  and  $\sqrt{J}$  are coprime, then  $I$  and  $J$  are coprime.*

**Proposition 3.4.** *Let  $R$  be a ring. Then every idempotent of  $R/\mathfrak{N}$  lifts to some idempotent of  $R$ . In fact, the lifting is unique.*

#### 3.2 Localization

**Proposition 3.5.** *Let  $R$  be a ring. Then  $R = \bigcap_{\mathfrak{m} \in \text{Max}(R)} R_{\mathfrak{m}}$ .*

#### 3.3 Modules

**Proposition 3.6.** *Let  $R$  be a noetherian ring. Then submodule of finitely generated  $R$ -module is still finitely generated.*

#### 3.4 Algebras

##### 3.4.1 finiteness

**Proposition 3.7.** *Let  $B$  be a finite  $A$ -algebra with homomorphism  $\varphi : A \rightarrow B$ . Assume  $\mathfrak{m}$  is a maximal ideal in  $A$ . Then  $B_{\mathfrak{m}} \cong \bigoplus_i B_{\mathfrak{p}_i}$ , where  $\mathfrak{p}_i$  varies over all prime ideals of  $B$  lying over  $\mathfrak{m}$ .*

##### 3.4.2 integrality

**Proposition 3.8.** *Let  $L/K$  be an algebraic field extension,  $A \subseteq B$  and  $\text{Frac } A = K$ . Assume  $B$  is the integral closure of  $A$  in  $L$ . Then  $\text{Frac } B = L$ .*

**Proposition 3.9.** *Let  $L/K$  be an algebraic field extension,  $A \subseteq B$  and  $\text{Frac } A = K$ . Assume  $B$  is the integral closure of  $A$  in  $L$ . Then  $B \otimes_A K = L$ .*

**Proposition 3.10.** *Let  $A \subseteq B$  be an integral extension with  $A$  integral domain. Then  $A$  is a field if and only if  $B$  is a field.*

### 3.4.3 flatness

**Proposition 3.11.** *Let  $f : A \rightarrow B$  be an etale homomorphism i.e. flat and unramified. If  $A$  is regular, then  $B$  is regular.*

## 3.5 Dimension theory

**Proposition 3.12.** *Let  $R$  be an artinian integral domain. Then  $R$  is a field.*

## 3.6 Valuation rings

**Proposition 3.13.** *Let  $R$  be a valuation ring,  $a, b \in R$ . Then the ultrametric inequality  $v(a + b) \geq \min\{v(a), v(b)\}$  takes equality whenever  $v(a) \neq v(b)$ .*

**Proposition 3.14.** *Let  $R$  be a local ring,  $\mathfrak{m} \subseteq R$  maximal ideal. If  $\dim_{R/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2) = 1$ , then  $\mathfrak{m}$  is principal.*

**Proposition 3.15.** *Let  $R$  be UFD. Then  $R$  is integrally closed.*

**Proposition 3.16.** *Let  $B \in \text{Alg}_A$  with ring homomorphism  $f : A \rightarrow B$ . For  $\mathfrak{p} \in \text{Spec } A$ , if  $\mathfrak{p}^{ec} = \mathfrak{p}$ , then  $\mathfrak{p} \in \text{im}(f^*)$ .*

**Proposition 3.17.** *Let  $A$  be a graded ring and  $M, N, L$  graded  $A$ -modules. Then  $\text{Hom}(M \otimes_A N, L) \cong \text{Hom}(M, \text{Hom}(N, L))$ .*

**Proposition 3.18.** *Let  $\varphi : A \rightarrow B$  be a homomorphism between noetherian rings. Then for each minimal prime ideal  $\mathfrak{p}$  containing  $\ker(\varphi)$  such that  $\varphi(\mathfrak{p}B) \neq B$ , we can find  $\mathfrak{P} \in \text{Spec } B$  lying over  $\mathfrak{p}$ .*

**Corollary 3.1.** *Let  $\varphi : A \rightarrow B$  be a homomorphism between noetherian rings. Then for each prime ideal  $\mathfrak{p}$  containing  $\ker(\varphi)$  such that  $\varphi(\mathfrak{p}B) \neq B$ , we can find  $\mathfrak{P} \in \text{Spec } B$  lying over  $\mathfrak{p}$ .*

## 4 Linear Algebra 线性代数

**Proposition 4.1.** *If  $U, V$  are subspaces of dimension  $r, s$  of a vector space of  $W$  of dimension  $n$ , then  $U \cap V$  is a subspace of dimension more than  $r + s - n$ .*

## 5 Representation Theory 表示论

**Theorem 5.1 (Burnside  $p^a q^b$  Theorem).**  *$G$  is a finite group of order  $p^a q^b$ , where  $p, q$  are prime numbers. Then  $G$  is solvable group.*

**Theorem 5.2 (Clifford Theorem).**  *$G$  is a finite group,  $N \trianglelefteq G$  a normal subgroup. Assume that  $\chi \in \text{Irr}(G)$  and  $\varphi \in \text{Irr}(N)$  satisfy that  $\varphi$  is a direct summand of  $\text{Res}_N^G(\chi)$ . Then  $\text{Res}_N^G(\chi) = e \sum_{i=1}^n \varphi^{g_i}$ , where  $e \in \mathbb{N}$ ,  $\varphi^{g_i}$  is given by  $x \mapsto \varphi(g_i x g_i^{-1})$  and  $g_1, \dots, g_n$  is a transversal of  $N$  in  $G$ .*

## 6 Algebraic Geometry 代数几何

### 6.1 Classic examples

**Proposition 6.1.** *Let  $\mathcal{L}$  be an invertible sheaf on projective line  $\mathbb{P}_k^1$ . Then  $\mathcal{L}$  must be isomorphic to  $\mathcal{O}_{\mathbb{P}_k^1}(n)$  for some  $n$ .*

### 6.2 Spectrum

**Proposition 6.2.** *Let  $\varphi : A \rightarrow B$  be a ring homomorphism. Assume that  $f : \text{Spec } B \rightarrow \text{Spec } A$  is the corresponding morphism of spectrum. Then  $\text{im } f \subseteq V(\ker \varphi)$ .*

**Corollary 6.1.** *Let  $\varphi : A \rightarrow B$  be a homomorphism between noetherian rings. Assume that  $f : \text{Spec } B \rightarrow \text{Spec } A$  is the corresponding morphism of spectrum. Then  $\text{im } f = \{\mathfrak{p} \in V(\ker \varphi) \mid \varphi(\mathfrak{p})B \neq B\}$ .*

**Corollary 6.2.** *Let  $\varphi : A \rightarrow B$  be a ring homomorphism. Assume that the corresponding morphism  $f : \text{Spec } B \rightarrow \text{Spec } A$  is surjective. Then  $\ker \varphi \subseteq \mathfrak{N}_A$ .*

**Proposition 6.3.** *Let  $\varphi : A \rightarrow B$  be a surjective ring homomorphism. Then the corresponding morphism  $f : \text{Spec } B \rightarrow \text{Spec } A$  is injective.*

**Proposition 6.4.** *Let  $\varphi : A \rightarrow B$  be a ring homomorphism. Assume that  $f : \text{Spec } B \rightarrow \text{Spec } A$  is the corresponding morphism of spectrum. Then  $f$  is a closed immersion if and only if  $\varphi$  is surjective.*

### 6.3 Properties of morphisms

**Proposition 6.5.** *Let  $X$  be a reduced scheme and  $Y$  be a separated scheme. Assume  $f$  and  $g$  are two morphisms from  $X$  to  $Y$ , agreeing on an open dense subset of  $X$ . Then  $f = g$ .*

**Remark 6.1.** *With this proposition, the equivalence classes of rational classes from reduced scheme to separated scheme are well behaved. This is why we always assume schemes are reduced and separated when we consider birational maps.*

### 6.4 Dimension theory

**Proposition 6.6.** *Let  $X$  be an irreducible affine scheme. Then for any nonempty open subset  $U \subseteq X$ , we have that  $\dim U = \dim X$ .*

**Remark 6.2.** *This is not true for general topological space.*

**Proposition 6.7.** *Let  $X$  be a noetherian topological space,  $Y \subseteq X$  constructible set. Then  $\dim Y = \dim \overline{Y}$ .*

### 6.5 Universal injectivity

**Proposition 6.8.** *Let  $f : \operatorname{Spec} L \rightarrow \operatorname{Spec} K$  be a morphism of schemes. Then  $f$  is universally injective if and only if field extension  $L/K$  is purely inseparable.*

**Proposition 6.9.** *Let  $f : X \rightarrow Y$  be a morphism of schemes. Then  $f$  is universally injective if and only if  $\Delta_{X/Y} : X \rightarrow X \times_Y X$  is surjective.*

### 6.6 Exceptional locus

**Proposition 6.10.** *Let  $f : X \rightarrow Y$  be a birational morphism between reduced and separated schemes,  $g : Y \dashrightarrow X$  the birational inverse of  $f$ . Then  $\{x \in X \mid g \text{ is not defined at } f(x)\}$  and  $X \setminus U_X$  are the same set, denoted by  $\operatorname{Ex}(f)$ , where  $U_X$  denotes the maximal open dense subset of  $X$  such that  $f|_{U_X}$  is an isomorphism.*

**Proposition 6.11.** *Let  $f : X \dashrightarrow Y$  be a birational map between reduced, noetherian and separated schemes,  $g : Y \dashrightarrow X$  the birational inverse of  $f$ . Then for any prime divisor  $D$  on  $Y$  not contained in  $\operatorname{Sing} Y$ , the generic point  $\eta$  of  $D$  is contained in  $U_Y$ .*

### 6.7 Ramification theory

**Proposition 6.12.** *Let  $f : X \rightarrow Y$  be a finite morphism between smooth algebraic curves of degree  $d$ . Assume  $P_1, \dots, P_r$  are all preimages of  $Q \in Y$ . Then*

$$\sum_{i=1}^r e_{P_i} = d \quad (1)$$

## 7 Homological Algebra

**Proposition 7.1.** *Let  $\mathcal{A}$  be an abelian category. Assume  $I^\bullet$  be an acyclic bounded-below complex of injective objects. Then  $I^\bullet$  is contractible i.e.  $\operatorname{id}_{I^\bullet}$  is null-homotopic. Dually, acyclic bounded-above complex of projective objects is contractible.*

## 8 Proofs

### 8.1 Abstract Algebra 抽象代数

**Proposition 1.1.**  $b = b1 = bac = 1c = c$  □

**Proposition 1.2.** If there exists  $z \notin R$ , while  $z \in \cap_{\alpha} R_{\mathfrak{m}_{\alpha}}$ , consider the ideal  $I = \{x \in R \mid xz \in R\}$ . Since  $1 \notin I$ , we have  $I$  is a proper ideal of  $R$ , thus there is a maximal ideal  $\mathfrak{m}$  s.t.  $I \subset \mathfrak{m}$ . However, it is obvious that  $z \notin R_{\mathfrak{m}}$ , contradiction! □

**Proposition 1.1.** See in the website, [Schur's Theorem on commutator subgroup](#). □



**Proposition 1.3.** Take normal cyclic subgroup  $H$  s.t. the index of  $H$  is minimal. Consider the centralizer  $C$  of  $H$ , and use Theorem 1.1 to show  $C = H$ . Finally, suppose there is an element  $g \notin H$ , try to make contradiction.  $\square$

**Proposition 1.4.** Consider the double coset for  $H$ .  $\square$

**Proposition 1.5.** Abelian case by Fundamental Theorem of Finitely Generated Abelian Groups and nonabelian case by Sylow Theorem.  $\square$

**Proposition 1.6.** By induction.  $\square$

**Proposition 1.7.** It suffices to prove the case for  $p$ -group.  $G$  is a  $p$ -group of order  $p^n$ , denote the number of maximal subgroups of  $G$  by  $m_G$ . By induction, we assume that  $m_G \equiv 1 \pmod{p}$  for  $\forall 1 \leq n \leq k$ .

Now consider the  $k+1$  case. If  $m_G = 1$ , then done. Otherwise, assume  $P, P'$  are two different maximal subgroups of  $G$ . Note that by Proposition 1.6, we have  $P \trianglelefteq G$ ,  $P' \trianglelefteq G$  and  $PP' = G$ . In addition,  $PP'/P \cong P'/P \cap P'$ . Thus  $\#(P \cap P') = p^{k-1}$ . For each maximal subgroup  $H$  of  $P, P'$  such that  $P' \cap P = H$  one-to-one corresponds to a subgroup of  $G/H$  of order  $p$  except  $P/H$ . The correspondence gives a bijection. Thus  $m_G \equiv 1 + \sum_{H \text{ maximal subgroup of } P} (m_{G/H} - 1) \equiv 1 \pmod{p}$ .  $\square$

**Proposition 1.8.** Obviously,  $G$  has no proper subgroup of index less than 5. Similarly, by Sylow's Theorem, the number of Sylow 3-groups and Sylow 5-groups are 10 and 6 respectively. And the number of Sylow 2-groups is 5 or 15. It is obvious to show  $G \cong A_5$  in the first case. For the other case, there must exists two different Sylow 2-groups  $P$  and  $Q$  such that  $P \cap Q \neq \{1\}$ . Assume that  $\langle a \rangle = P \cap Q$ , then  $o(a) = 2$ . Consider the order of  $C_G(a)$ .  $\square$

**Proposition 1.9.** For any  $g \in G$ ,  $g^{-1}Pg \subseteq g^{-1}Ng = N$ . Thus  $g^{-1}Pg$  is still some Sylow  $p$ -group of  $N$ . By Sylow's Theorem, there exists  $n \in N$  such that  $g^{-1}Pg = n^{-1}Pn$ . Then  $g = (gn^{-1})n$ , where  $gn^{-1} \in N_G(P)$ .  $\square$

## 8.2 Topology 拓扑

**Proposition 2.1.** " $\Rightarrow$ ": obviously. " $\Leftarrow$ ": consider  $U_\alpha \setminus S$   $\square$

**Proposition 2.2.** Suppose  $V$  is a maximal irreducible subspace of  $X$ . Consider the closure of  $V$ .  $\square$

**Proposition 2.3.** Assume there is infinitely many irreducible components. Consider the descending chain of closed subsets

$$V_1 \supseteq V_1 \cap V_2 \supseteq \cdots \supseteq V_1 \cap \cdots \cap V_n \supseteq \cdots \quad (2)$$

where  $V_i$  are distinct irreducible components.  $\square$

### 8.2.1 dimension theory

**Proposition 2.4.** Suppose that  $U$  is not dense, then there exist  $x \in X$  and open neighbourhood  $W \ni x$  such that  $W \cap U = \emptyset$ , contradicting to irreducibility. Hence  $U$  is dense in  $X$ .  $\square$

## 8.3 Commutative Algebra 交换代数

### 8.3.1 Radical

**Proposition 3.1.** See in the Atiyah.  $\square$

**Proposition 3.2.** See in the Atiyah.  $\square$

**Proposition 3.3.**  $r(I \cup J) = r(r(I) \cup r(J)) = r(1) = (1)$   $\square$

**Proposition 3.4.** Assume  $\bar{x}$  is the idempotent in  $A/\mathfrak{N}$ , we have  $x^2 - x$  is nilpotent. Thus  $\exists n \in \mathbb{N}_+$  s.t.  $x^n(1 - x)^n = 0$ . As  $(x)$  and  $(1 - x)$  are coprime, we have  $r(x)$  and  $r(1 - x)$  are coprime. Note that  $r(x^n) = r(x)$  and  $r((1 - x)^n) = r(1 - x)$ , by Proposition 3.3, get  $(x^n)$  and  $((1 - x)^n)$  are coprime. By Chinese Remainder Theorem, we have

$$A = A/(0) = A/(x^n(1 - x)^n) \cong A/(x^n) \times A/((1 - x)^n) \quad (3)$$

Take  $e$  be the preimage of  $(0, 1) \in A/(x^n) \times A/((1 - x)^n)$ ,  $e$  is the lifting of  $x$ .  $\square$

**Proposition 3.5.** Consider natural map  $f : R \longrightarrow \bigcap_{\mathfrak{m} \in \text{Max}(R)} R_{\mathfrak{m}}$ . Want to show that  $f$  is surjective. It suffices to prove that  $f_{\mathfrak{m}} : R_{\mathfrak{m}} \longrightarrow (\bigcap_{\mathfrak{m} \in \text{Max}(R)} R_{\mathfrak{m}})_{\mathfrak{m}}$  is surjective for each maximal ideal  $\mathfrak{m}$ .  $\square$

### 8.3.2 Modules

**Proposition 3.6.** Only need to prove for  $A^n$ -module. Consider the exact sequence

$$0 \longrightarrow A^{n-1} \longrightarrow A^n \longrightarrow A \longrightarrow 0 \quad (4)$$

from  $A^n$ -module  $M$ , we can get  $N \subseteq A^{n-1}$  module and  $I \subseteq A$  ideal. And from  $N$  and  $I$ , we can generate  $M$ .  $\square$

### 8.3.3 Algebras

#### finiteness

**Proposition 3.7.** i don't know  $\square$

#### integrality

**Proposition 3.8.** Since  $L/K$  is algebraic, for all  $\alpha \in L$ , there is a minimal polynomial  $x^n + \frac{a_1}{a'_1}x^{n-1} + \dots + \frac{a_n}{a'_n}$ , where all  $a_i$  and  $a'_i$  are in  $A$ . Take the multiplication  $a$  of all dominators, then  $a\alpha$  is a root of  $x^n + \frac{a_1a}{a'_1}x^{n-1} + \dots + \frac{a_na^n}{a'_n}$  with all coefficients in  $A$ . By definition, we know  $a\alpha \in B$  and hence  $\alpha \in \text{Frac } B$ . In conclusion,  $\text{Frac } B = L$ .  $\square$

**Proposition 3.9.** By Proposition 3.8, we only need to show that for all  $b \in B$ ,  $b^{-1} \in B \otimes_A K$ . In fact,  $B \otimes_A K = S^{-1}B$  where  $S = A \setminus \{0\}$ . And for all  $b \in B$ , there is a minimal polynomial  $x^n + a_1x^{n-1} + \cdots + a_n$  with coefficients in  $A$ . By definition, we know  $a_n \neq 0$ . Denote  $c = b^{n-1} + a_1b^{n-2} + \cdots + a_{n-1}$ , we get that  $bc = -a_n \in A$  so that  $b^{-1} = \frac{c}{-a_n}$ .  $\square$

**Proposition 3.10.** When  $B$  is a field, since  $A$  is a subring of  $B$ ,  $A$  is also a field. Conversely, assume  $A$  is a field. As  $B$  is integral over  $A$ , for all nonzero  $b \in B$ , there is a monic polynomial  $x^n + a_1x^{n-1} + \cdots + a_n$  of  $b$  over  $A$  with  $a_n \neq 0$ . Then  $b(b^{n-1} + a_1b^{n-2} + \cdots + a_{n-1}) \in A^\times$  so that  $b$  is invertible. Conclude that  $B$  is a field.  $\square$

flatness

**Proposition 3.11.** i don't know.  $\square$

### 8.3.4 Dimension theory

**Proposition 3.12.** For any  $x \neq 0 \in R$ , consider the descending chain of ideals

$$(x) \supseteq (x^2) \supseteq \cdots \supseteq (x^n) \supseteq \cdots \quad (5)$$

By Artinian property, we can get that  $x$  is invertible.  $\square$

### 8.3.5 Valuation rings

**Proposition 3.13.** We may assume  $v(a) < v(b)$  so that  $v(a+b) \geq v(a)$ . Again by the ultrametric inequality,  $v(a) \geq \min\{v(a+b), v(-b)\}$ . Since  $v(-b) = v(b) > v(a)$ , we get  $v(a) \geq v(a+b)$ , done!  $\square$

**Proposition 3.14.** Assume that  $\mathfrak{m}/\mathfrak{m}^2 = R/\mathfrak{m}\bar{x}$ . Consider the correspondence between ideals of  $R/\mathfrak{m}^2$  and ideals of  $R$  containing  $\mathfrak{m}^2$ . Get  $\mathfrak{m} = (x) + \mathfrak{m}^2$ . By Nakayama's Lemma, we have  $\mathfrak{m} = (x)$ .  $\square$

**Proposition 3.15.** For any  $\frac{a}{b} \in \text{Frac}(R) \setminus R$  integral over  $R$ , assume that  $\gcd(a, b) = 1$ . There exists a monic polynomial  $x^n + a_1x^{n-1} + \cdots + a_n$  where  $a_i \in R$  s.t.  $(\frac{a}{b})^n + a_1(\frac{a}{b})^{n-1} + \cdots + a_n = 0$ . Thus  $a^n = -b(a_1a^n + \cdots + a_nb^{n-1})$ , get  $b \mid a^n$  which contradicts to our assumption,  $\gcd(a, b) = 1$ .  $\square$

**Proposition 3.16.** Consider  $f : A_{\mathfrak{p}} \rightarrow B_{\mathfrak{p}}$ . Then the maximal ideal of  $B_{\mathfrak{p}}$  lies over  $\mathfrak{p}$ .  $\square$

**Proposition 3.17.** By definition.  $\square$

**Proposition 3.18.** For any minimal prime ideal  $\mathfrak{p}$  of  $A$  containing kernel, consider set  $V = V(\varphi(\mathfrak{p}))$  in  $\text{Spec } B$ , which by assumption is nonempty. As  $A, B$  are both noetherian,  $V$  is finite. Suppose that preimages of all  $\mathfrak{P} \in V(\varphi(\mathfrak{p}))$  are not  $\mathfrak{p}$ , then each  $\mathfrak{P}_i$  gives an  $a_i \notin \mathfrak{p}$ . Set  $a = \prod_i a_i$ , then  $a \notin \mathfrak{p}$  and  $\varphi(a) \in \cap_i \mathfrak{P}_i$  so that  $\varphi(a) \in \sqrt{\varphi(\mathfrak{p})}$ . Note that  $\varphi^{-1}(\varphi(\mathfrak{p})) = \mathfrak{p}$ , we get  $a \in \mathfrak{p}$ , contradicton.  $\square$

**Corollary 3.1.** Consider ring homomorphism  $\tilde{\varphi} : A/\mathfrak{p} \rightarrow B/\varphi(\mathfrak{p})B$ . Then apply Proposition 3.18, done!  $\square$

## 8.4 Linear Algebra 线性代数

**Proposition 4.1.** Consider the maximal linearly independent system □

## 8.5 Representation Theory 表示论

**Theorem 5.1.** See in the note. □

**Theorem 5.2.** Note that  $\chi^{g_i} = \chi$ , we have

$$\begin{aligned} \langle \text{Res}_N^G(\chi), \varphi^{g_i} \rangle &= \langle \text{Res}_N^G(\chi^{g_i}), \varphi^{g_i} \rangle \\ &= \langle \text{Res}_N^G(\chi)^{g_i}, \varphi^{g_i} \rangle \\ &= \langle \text{Res}_N^G(\chi), \varphi \rangle \\ &= e \end{aligned}$$

For the proof of showing  $\varphi^{g_i}$  are all the direct summand of  $\text{Res}_N^G(\chi)$ , see in the book Representation Theory of Finite Group Extensions. □

## 8.6 Algebraic Geometry

### 8.6.1 Classic examples

**Proposition 6.1.** See in the website, [invertible sheaves of the projective line](#). □

### 8.6.2 Spectrum

**Proposition 6.2.** It is clear that for all  $\mathfrak{p} \in \text{Spec } B$ ,  $\ker \varphi \subseteq \varphi^{-1}(\mathfrak{p})$  so that  $f(\mathfrak{p}) \in V(\ker \varphi)$ . Thus  $\text{im } f \subseteq V(\ker \varphi)$  □

**Corollary 6.1.** Immediately comes from Proposition 6.2 and Corollary 3.1. □

**Corollary 6.2.** Immediately comes from Proposition 6.2. □

**Proposition 6.3.** For different  $\mathfrak{p}_1$  and  $\mathfrak{p}_2$  in  $\text{Spec } B$ , there must exists  $a \in \mathfrak{p}_1 \setminus \mathfrak{p}_2$  so that  $\varphi^{-1}(a) \subseteq \varphi^{-1}(\mathfrak{p}_1) \setminus \varphi^{-1}(\mathfrak{p}_2)$  and hence  $f(\mathfrak{p}_1) \neq f(\mathfrak{p}_2)$ . □

**Proposition 6.4.** □

### 8.6.3 Properties of morphisms

**Proposition 6.5.** Since the problem is locally, we may assume  $X = \text{Spec } A$  and  $Y = \text{Spec } B$  are both affine. Consider the following commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow g & & \downarrow \\ Y & \longrightarrow & \text{Spec } \mathbb{Z} \end{array} \quad (6)$$

By universal property of fibered product, we get a lifting  $(f, g) : X \rightarrow Y \times Y$ . Note that the set  $\{x \in X \mid f(x) = g(x)\}$  is the preimage of  $\Delta_Y$  under  $(f, g)$ , which is a closed subset

of  $X$  since  $Y$  is separated. As  $f$  and  $g$  agree on an open dense subset  $U$  of  $X$ , we get  $U \subset \{x \in X \mid f(x) = g(x)\}$  and hence  $\{x \in X \mid f(x) = g(x)\} = X$ .

Now we have two ring homomorphisms  $f^\# : B \rightarrow A$  and  $g^\# : B \rightarrow A$  satisfying that for all  $\mathfrak{p} \in \text{Spec } A$ ,  $f(\mathfrak{p}) = g(\mathfrak{p})$ . Remains to show  $f^\# = g^\#$ . As  $X$  is reduced, it suffices to show that for all  $b \in B$ ,  $V(f^\#(b) - g^\#(b)) = X$ . Since  $U$  is dense in  $X$ , it suffices to show that  $U \subseteq V(f^\#(b) - g^\#(b))$ .

In fact, since for all  $\mathfrak{p} \in U$ , we can take an affine open neighbourhood  $D(s) \subseteq U$ . Then  $B \xrightarrow{f^\#} A \rightarrow A_s$  and  $B \xrightarrow{g^\#} A_s$  are same. Hence we get  $\frac{f^\#(b)}{1} = \frac{g^\#(b)}{1}$  in  $A_s$  and  $(f^\#(b) - g^\#(b))s^n = 0$  for some  $n$ . As  $s \notin \mathfrak{p}$ ,  $f^\#(b) - g^\#(b) \in \mathfrak{p}$  and we are done!  $\square$

#### 8.6.4 Dimension Theory

**Proposition 6.6.** See in the website, [dimension of irreducible affine variety is same as any open subset](#).  $\square$

**Proposition 6.7.** Since constructible can be written as finite union of locally closed subsets, assume that  $Y = \cap_{i=1}^n U_i \cap V_i$ . In particular, by taking irreducible components of each  $V_i$ , we may assume that  $V_i$  are irreducible. Then  $\dim \bar{Y} \geq \dim Y \geq \max_i \{\dim(U_i \cap V_i)\}$ . Note that  $\bar{Y} = \cup_{i=1}^n \bar{U}_i \cap \bar{V}_i$  and by Proposition 6.6, we immediately get

$$\dim \bar{Y} = \max_i \{\dim(\bar{U}_i \cap \bar{V}_i)\} = \max_i \{\dim V_i\} = \max_i \{\dim(U_i \cap V_i)\} \quad (7)$$

Thus  $\dim Y = \dim \bar{Y}$ .  $\square$

#### 8.6.5 Universal injectivity

**Proposition 6.8.** I don't know  $\square$

**Proposition 6.9.** As composition of projection map and  $\Delta_{X/Y}$  is identity map of  $X$ , if  $f$  is universally injective, then projection map is injective and hence  $\Delta_{X/Y}$  is surjective.

Conversely, for any morphism  $Z \rightarrow Y$ , it suffices to show that  $\Delta_{X \times_Y Z/Z}$  is surjective. Note that there is a fibered diagram

$$\begin{array}{ccc} X \times_Y Z & \xrightarrow{\Delta_{X \times_Y Z/Z}} & X \times_Y Z \times_Z X \times_Y Z \cong X \times_Y X \times_Y Z \\ \downarrow & & \downarrow \\ X & \xrightarrow{\Delta_{X/Y}} & X \times_Y X \end{array} \quad (8)$$

Then as base change preserves surjectivity, we get  $\Delta_{X \times_Y Z/Z}$  is surjective.  $\square$

#### 8.6.6 Exceptional locus

**Proposition 6.10.** Obviously,  $\{x \in X \mid g \text{ is not defined at } f(x)\} \subseteq X \setminus U_X$ . We only need to show the other containment. Assume  $g$  is defined at  $f(x)$ . Then by definition of rational map, there are open neighbourhoods  $U \ni x$  and  $V \ni x$  such that  $f(U) \subseteq V$  and  $g$  is defined on  $V$ .

As  $U_X$  is open dense subset of  $X$ ,  $U_X \cap U$  is nonempty and dense in  $U$ . Hence  $g \circ f|_U$  and  $U \hookrightarrow X$  agree on  $U_X \cap U$ . By Proposition 6.5,  $g \circ f|_U$  is just the open immersion.

Hence we can replace  $V$  by  $V \cap g^{-1}(U)$ , which is still an open neighbourhood of  $f(x)$ . Applying same argument for  $f \circ g|_V$ , we conclude that  $U \subseteq U_X$  and  $V \subseteq U_Y$ , done!  $\square$

**Proposition 6.11.** As  $D$  is not contained in  $\text{Sing } Y$ ,  $Y$  is regular at  $\eta$  so that  $\mathcal{O}_{Y,\eta}$  is a DVR. Assume  $Z$  is an irreducible component of  $Y$  such that  $D \subseteq Z$  and  $\xi$  is the generic point of  $Z$ , then  $\mathcal{O}_{Y,\xi} = \text{Frac}(\mathcal{O}_{Y,\eta})$ .

Since  $U_Y$  is dense in  $Y$ ,  $\xi \in U_Y$  so that  $g$  induces a map  $\text{Spec } \mathcal{O}_{Y,\xi} \rightarrow U_X$ . And we have a commutative diagram

$$\begin{array}{ccc} \text{Spec } \mathcal{O}_{Y,\xi} & \longrightarrow & U_X \\ \downarrow & \nearrow & \downarrow f \\ \text{Spec } \mathcal{O}_{Y,\eta} & \longrightarrow & Y \end{array} \quad (9)$$

Applying valuative criterion of properness, there exists unique lifting  $\text{Spec } \mathcal{O}_{Y,\eta} \rightarrow X$ . Hence there exists  $\delta \in U_X$  such that  $f(\delta) = \eta$ . By definition, we know  $\eta \notin \text{Ex}(g)$ .  $\square$

### 8.6.7 Ramification theory

**Proposition 6.12.** As  $f$  is a finite morphism of degree  $d$ , the stalk  $(f_*\mathcal{O}_X)_Q$  is a free  $\mathcal{O}_{Y,P}$ -module of rank  $d$ . Tensoring by  $k(Q)$ , we get a  $d$ -dimensional vector space over  $k(Q)$ . Note that by Proposition 3.7, we have that

$$\begin{aligned} (f_*\mathcal{O}_X)_Q \otimes_{\mathcal{O}_{Y,Q}} k(Q) &\cong (\oplus_{i=1}^r \mathcal{O}_{X,P_i}) \otimes_{\mathcal{O}_{Y,Q}} k(Q) \\ &\cong \oplus_{i=1}^r \mathcal{O}_{X,P_i} / \mathfrak{m}_Q \mathcal{O}_{X,P_i} \\ &\cong \oplus_{i=1}^r \mathcal{O}_{X,P_i} / \mathfrak{m}_{P_i}^{e_{P_i}} \end{aligned} \quad (10)$$

Counting dimension, we get

$$d = \sum_{i=1}^r e_{P_i} \quad (11)$$

$\square$

## 8.7 Homological Algebra 同调代数

**Proposition 7.1.** Here we only prove for the injective version. As  $I^\bullet$  is acyclic, denote  $\ker \partial_n = Z_n$ . Then there are short exact sequences

$$0 \longrightarrow Z_n \longrightarrow I^n \longrightarrow Z_{n+1} \longrightarrow 0 \quad (12)$$

Since  $I^\bullet$  is bounded-below, we may assume for all negative  $i$ ,  $I^i = 0$ . Hence  $Z_2 = I^1$ . Consider the short exact sequence

$$0 \longrightarrow I^1 \longrightarrow I^2 \longrightarrow Z_3 \longrightarrow 0 \quad (13)$$

As  $I^1$  is injective, the exact sequence splits so that  $Z_3$  is a summand of  $I^2$  and hence injective. By induction, we get that all those short exact sequences split. Now we have projections

$p_n : I^n \rightarrow Z_n$  and inclusions  $s_{n+1} : Z_{n+1} \rightarrow I^n$ . Consider  $D_n := s_n \circ p_n : I^n \rightarrow I^{n-1}$ . In fact we have that

$$\partial \circ D_n + D_{n+1} \circ \partial = p_n + s_{n+1} = \text{id}_{I^n} \quad (14)$$

Hence  $I^\bullet$  is contractible. □